Domain-Specific Search: BM25 vs. Neural Retrieval on Technical Documents

Final Report | Information Retrieval | Group 10

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Repository: https://github.com/TODO/bm25-vs-neural

Status: PRELIMINARY: 325 annotator conflicts are not yet adjudicated. Every cs.CR effectiveness number below is shown as a range: the value with all unresolved conflicts at the lower grade, and with all of them at the higher grade. These two runs are a sensitivity check, not bounds (nDCG is not monotone in the grades). Exact bounds over all adjudications, and results for random adjudications, are given in Table 3. The SciFact numbers (Section 5.1) only validate the implementation and are not cs.CR results.

1 Introduction

Dense retrievers trained on general web data are often assumed to beat lexical retrieval. We test this on technical text where exact strings matter: computer-security abstracts from arXiv (cs.CR), which are full of identifiers (CVE numbers, protocol versions), homonyms (*oracle*, *poisoning*, *fingerprinting*) and conditions such as exclusions and date limits. We compare our identifier-aware BM25, two dense models used without fine-tuning (E5-base-v2 and SPECTER2), and their Reciprocal Rank Fusion hybrids. Research questions: **RQ1** Does a dense retriever beat BM25 overall? **RQ2** How does this depend on the query type? **RQ3** Does the science-specific SPECTER2 beat the general E5? **RQ4** Does a hybrid beat both of its components? **RQ5** How sensitive is BM25 to k_1, b and tokenisation?

2 Related work

BM25 [1] remains a strong zero-shot baseline: in BEIR [2] it beat many dense models outside their training domain. E5 [3] is trained contrastively on large weakly supervised pair data and then fine-tuned. SPECTER2 [4] is trained on scientific citation data, with task adapters for proximity and ad-hoc search. RRF [5] fuses ranked lists using ranks only. Bruch et al. [6] compare RRF with convex score combination. Dense models are known to handle negation poorly [7]. Our design for significance tests follows Smucker et al. [8]; for assessor disagreement it follows Voorhees [9].

3 Data, queries and systems

Corpus. We use all 39,613 arXiv papers with cs.CR as primary or cross-listed category and a first version in 2016–2025. Withdrawn papers and duplicate titles are removed and LaTeX is cleaned (SHA-256 `fda391ecc5b6...`). A document is its title plus abstract. **Queries.** 50 queries, 10 per category: exact identifier (e.g. "Spectre variant 1 bounds check bypass (CVE-2017-5753)"), homonym ("padding oracle attacks on CBC mode"), negation ("anonymous communication systems that are *not* based on Tor"), time constraint ("attacks on TLS 1.3 published after its 2018 standardisation") and paraphrase ("copying a deployed model by repeatedly querying its prediction interface"). Each query has a description and written relevance criteria. **Systems.** (1) BM25 (k_1=1.2, b=0.75) with an identifier-aware analyzer. It keeps `cve-2021-44228` or `tls1.3` as single tokens and also indexes their parts; Lucene's stopwords are removed (including "not") and the Porter stemmer is applied. (2) E5-base-v2 and (3) SPECTER2 (proximity adapter for documents, ad-hoc query adapter for queries), both with exact cosine search. (4, 5) RRF (k=60, top 100 of each run) of BM25 with E5 and with SPECTER2. Nothing was tuned on our queries.

4 Evaluation protocol

Pooling and judging. The top 10 of all five runs form the pool: 1339 query–document pairs, about 27 per query. Every query was judged independently by two of the four authors, with the six annotator pairs spread evenly. The judging pages showed no system names or ranks, and showed dates only for time queries. Grades: 0 not relevant, 1 on topic but does not answer the query or breaks a condition, 2 relevant, 3 highly relevant. All 2,678 judgments were collected and audited: none is missing, duplicated or outside the pool. **Agreement.** Weighted Cohen's $\kappa = 0.760$ (linear) and 0.829 (quadratic); κ on binary relevance (grade ≥ 2) = 0.795; exact agreement 0.757. This meets the protocol's target of $\kappa \geq 0.6$. 325 pairs conflict: 21 are 0 vs 3, and 134 fall on opposite sides of the relevance line. By protocol, conflicts are adjudicated by the team before the qrels are frozen. **Metrics.** The primary metric is nDCG@10 with gain equal to the grade. P@5, MRR@10 and MAP@10 use grade ≥ 2. Unjudged documents count as non-relevant; Hole@10 reports how often they occur. Our metric code matches `trec_eval` (via `pytrec_eval`) on every query of every run, with a maximum difference of 3×10^{-16}. **Significance.** Paired randomisation tests over queries (100,000 permutations) on nDCG@10 for all 10 system

Table 1: Effectiveness on the 50 cs.CR queries. PRELIMINARY: each cell gives the value with all 325 unresolved conflicts at the lower grade – at the higher grade (a sensitivity check, not bounds; see Table 3). Binary metrics use grade ≥ 2; Hole@10 = share of top-10 documents never judged.

System	nDCG@10	nDCG@5	P@5	MRR@10	MAP@10	Hole@10
BM25	0.664–0.724	0.627–0.712	0.460–0.560	0.560–0.677	0.285–0.321	0.0%
E5	0.670–0.744	0.631–0.727	0.496–0.608	0.646–0.712	0.319–0.356	0.0%
SPECTER2	0.506–0.619	0.487–0.632	0.348–0.488	0.537–0.680	0.187–0.243	0.0%
BM25+E5	0.699–0.766	0.660–0.752	0.484–0.592	0.629–0.715	0.313–0.354	0.0%
BM25+SPECTER2	0.643–0.720	0.622–0.722	0.468–0.576	0.575–0.697	0.271–0.307	0.0%

Table 2: Paired randomisation tests on nDCG@10 over the 50 queries (100,000 permutations; Holm–Bonferroni over all 10 pairs; $\alpha = 0.05$). PRELIMINARY: ranges over the lower and upper fallback runs; 'depends' = significant in only one of them. Robustness to adjudication: Table 3.

Comparison	ΔnDCG@10	p_{Holm}	Significant
BM25 − E5	−0.020 to −0.006	1.000	no
BM25 − SPECTER2	+0.105 to +0.158	0.001–0.023	yes
BM25 − BM25+E5	−0.042 to −0.035	0.173–0.345	no
BM25 − BM25+SPECTER2	+0.005 to +0.021	1.000	no
E5 − SPECTER2	+0.125 to +0.164	<0.001	yes
E5 − BM25+E5	−0.029 to −0.021	0.624–1.000	no
E5 − BM25+SPECTER2	+0.025 to +0.027	1.000	no
SPECTER2 − BM25+E5	−0.193 to −0.147	<0.001	yes
SPECTER2 − BM25+SPECTER2	−0.137 to −0.101	<0.001	yes
BM25+E5 − BM25+SPECTER2	+0.046 to +0.056	0.074–0.231	no

pairs, with Holm–Bonferroni correction. Per-category results come with 95% bootstrap CIs and are exploratory, since each category has only 10 queries.

5 Results

5.1 Validation of the implementation (SciFact)

This is not a cs.CR result. On BEIR SciFact (300 queries) with the same settings as Pyserini, our BM25 differs from Lucene by ΔnDCG@10 = -0.0009 and shares 98.0% of its top-10 documents. On the same collection E5 (0.727) beats BM25 (0.677) significantly (p_{Holm} = 0.023), while SPECTER2 (0.646) does not.

5.2 Main results on cs.CR

Tables 1 and 2 show that Hole@10 is 0 for every system: every document in every top 10 was judged. Table 3 shows how far adjudication can move the comparisons. We enumerated every way of resolving each conflict to one of its two submitted grades, which gives exact ranges of the mean differences. We also drew 500 random adjudications. **RQ1:** E5 and BM25 are statistically indistinguishable overall (difference 0.006–0.020 in favour of E5, $p_{\text{Holm}} = 1$). The sign of the difference depends on adjudication, and the difference is significant in none of the random adjudications. The data give no support for the claim that a general dense model beats BM25 on this collection. **RQ3:** SPECTER2 is the weakest system, significantly worse than BM25, E5 and both hybrids (0.105–0.164 below BM25 and E5). No adjudication can reverse the direction of these four differences, and they are significant in 99–100% of random adjudications. A deliberately adversarial adjudication can make them non-significant, however, so this conclusion is final only after the real adjudication. SPECTER2's citation-based training does not transfer to short ad-hoc queries. **RQ4:** BM25+E5 has the highest mean nDCG@10 under both fallback runs (0.699–0.766) and in 100% of random adjudications (exact range over all adjudications 0.657–0.798), but its gains over BM25 (+0.035–0.042) and E5 (+0.021–0.029) are not significant after correction. Fusing BM25 with SPECTER2 recovers most of SPECTER2's loss but does not beat BM25.

5.3 By query type (RQ2, exploratory)

Under both fallback runs (Table 4), BM25 leads on *exact identifiers* (0.708–0.825 vs. 0.583–0.722 for E5), while E5 leads on *paraphrases* (0.656–0.723 vs. 0.558–0.621) and *negation* (0.676–0.692 vs. 0.562–0.601). BM25+E5 is best on paraphrase and time queries. On homonyms, BM25, E5 and BM25+E5 are all high (above 0.77). With 10 queries per category the CIs overlap widely, so these are hypotheses for follow-up rather than findings. Within-category tests (Holm-corrected

Table 3: Sensitivity to adjudication. Exact range of each mean difference over every adjudication that picks one of the two submitted grades (40,269 per-query combinations enumerated); share of 500 random adjudications in which the pair is significant (Holm, 10 pairs); and the test on the single most adversarial adjudication.

Comparison	Exact range of ΔnDCG@10	Sign can flip	Sig. in random adj.	Worst case
BM25 − E5	−0.099 to +0.068	yes	0%	–
BM25 − SPECTER2	+0.001 to +0.250	no	99%	n.s.
BM25 − BM25+E5	−0.092 to +0.015	yes	7%	–
BM25 − BM25+SPECTER2	−0.064 to +0.088	yes	0%	–
E5 − SPECTER2	+0.020 to +0.262	no	100%	n.s.
E5 − BM25+E5	−0.080 to +0.035	yes	0%	–
E5 − BM25+SPECTER2	−0.067 to +0.118	yes	0%	–
SPECTER2 − BM25+E5	−0.283 to −0.046	no	100%	n.s.
SPECTER2 − BM25+SPECTER2	−0.202 to −0.032	no	100%	n.s.
BM25+E5 − BM25+SPECTER2	−0.021 to +0.122	yes	16%	–

Table 4: nDCG@10 by query category (10 queries each; exploratory). PRELIMINARY ranges as in Table 1.

System	Exact identifier	Homonym	Negation	Time constraint	Paraphrase
BM25	0.708–0.825	0.784–0.908	0.562–0.601	0.669–0.704	0.558–0.621
E5	0.583–0.722	0.774–0.877	0.676–0.692	0.645–0.722	0.656–0.723
SPECTER2	0.429–0.608	0.517–0.650	0.617–0.627	0.539–0.657	0.429–0.552
BM25+E5	0.660–0.789	0.771–0.898	0.590–0.624	0.715–0.764	0.726–0.787
BM25+SPECTER2	0.639–0.762	0.733–0.864	0.588–0.604	0.644–0.710	0.596–0.675

within each category) find only a few significant differences, none of them in both fallback runs: BM25+E5 over BM25 on paraphrases (lower bound), and SPECTER2 below each of the other four systems on homonyms (upper bound).

5.4 Error analysis

Conditions are ignored. On negation queries, 76–81% of BM25's top 10 are graded 1, mostly papers that break the exclusion. The figure is 59–63% for E5 and 68–74% for BM25+E5 (Table 5). BM25 removes "not" as a stopword. For "anonymous communication systems that are not based on Tor", nine of BM25's top 10 were graded 1 by both annotators, the grade the relevance criteria give to papers about Tor itself. Six of E5's top 10 were graded 2 or 3 by both annotators. On time queries, 39–58% of each system's top 10 is graded 1. Counting from first-version dates, 30–38% of each system's top 10 breaks the date condition; a rule-based date filter (stretch goal S2) removes all of these. **Paraphrases defeat lexical matching.** For "programs that lock a victim's files and demand payment to unlock them", BM25 ranks four non-relevant documents in its top five, while E5 returns ransomware papers that both annotators graded 2–3. **The two systems complement each other.** Of the relevant documents in the BM25 and E5 top 10s, 96–115 are found by both, 116–146 only by BM25 and 140–174 only by E5. This explains why fusion helps even though neither component is better overall. Full examples are in `deliverables/*/analysis/error_analysis.md`.

5.5 Efficiency

All numbers are measured from the run logs (Table 6). BM25 builds its index in 20 s on one CPU into 29 MB and answers in 1.6 ms. Each dense index is 116 MB (4× larger) and takes about 3 min of T4 GPU time. Dense queries take 17–30 ms, mostly query encoding; exact search alone takes 6–7 ms. BM25+E5 therefore buys its effectiveness at roughly 12× the query cost of BM25.

5.6 BM25 sensitivity (RQ5)

[Blocked until the qrels are frozen: `bm25_sweep.py` (with 5-fold cross-validation, so no tuning on the test queries is reported as a test score) and `tokenizer_ablation.py` need the corpus and run on Kaggle.]

6 Limitations

(1) The results are preliminary until the 325 conflicts are adjudicated. The system order is the same in 93% of random adjudications, but significance is guaranteed only for the real one. (2) There are 50 queries, 10 per category, so category effects are exploratory. (3) The pool contains only the top 10 of five systems. Hole@10 is 0 for these systems, but recall-oriented metrics, and any system outside the pool, would be biased. (4) No relevant document (grade ≥ 2) was

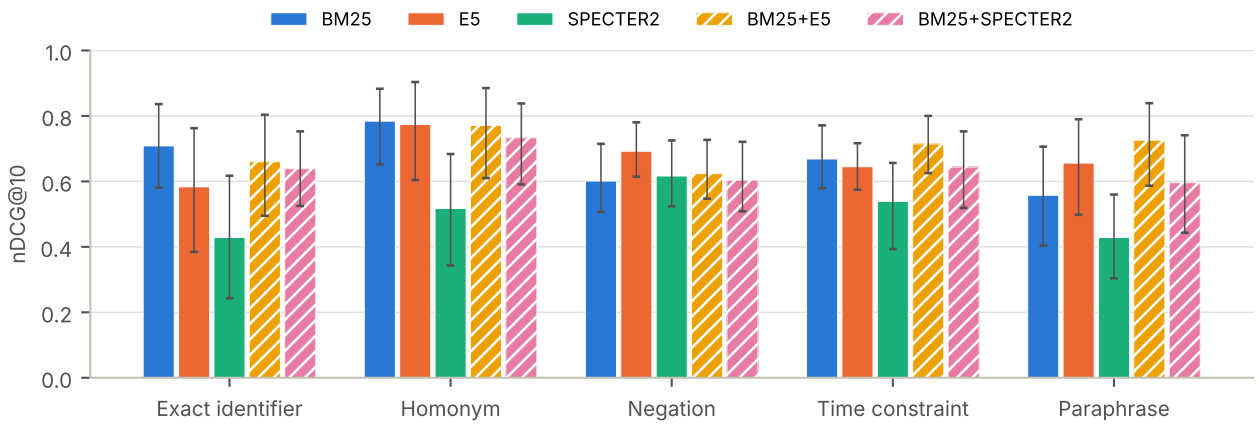


Figure 1: nDCG@10 per category with 95% bootstrap CIs, lower bound (hatched = hybrids). Exploratory.

Table 5: Share of each system's top 10 graded 1 (on topic but does not answer, or breaks the exclusion / date limit).

System	Exact identifier	Homonym	Negation	Time constraint	Paraphrase
BM25	0.36–0.39	0.10–0.12	0.76–0.81	0.52–0.58	0.12–0.16
E5	0.30–0.32	0.07–0.08	0.59–0.63	0.41–0.50	0.17–0.20
SPECTER2	0.27–0.35	0.10–0.24	0.61–0.74	0.39–0.43	0.26–0.31
BM25+E5	0.36–0.38	0.08–0.11	0.68–0.74	0.44–0.53	0.08–0.17
BM25+SPECTER2	0.35–0.43	0.08–0.14	0.73–0.78	0.43–0.49	0.16

found for one query (AES-GCM nonce reuse); it counts for every system, and removing it does not change the system order. (5) The authors wrote the queries and judged them. (6) The dense models may have seen arXiv abstracts during pre-training. (7) A grade of 1 earns nDCG gain, so systems partly get credit for documents that break a condition; an exploratory strict variant (no gain for grade 1, `sNDCG@10` in the result files) gives the same system order, except that BM25 and BM25+SPECTER2 tie under the upper bound.

7 Conclusions

On security abstracts, a general dense retriever used without fine-tuning does not beat a careful BM25 overall. The science-specific SPECTER2 is clearly worse than both. BM25 and E5 fail on different queries: BM25 on paraphrases and negation, E5 on exact identifiers. Their RRF fusion scores highest under every adjudication we sampled, but its gain over the better component is not significant with 50 queries, and it costs about 12× the query time of BM25. Every system largely ignores exclusions and date limits; handling them explicitly (S2) is more promising than switching models.

Contribution matrix

Member	Components
Surajsinh Patil	[fill in]
Tejas Mahajan	[fill in]
Swayam Kumud	[fill in]
Saksham Singhal	[fill in]
All	Query writing; relevance judging (about 670 judgments each); adjudication

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Table 6: Efficiency on the cs.CR corpus. Dense indexing = encoding the corpus on the GPU; dense query time includes encoding the query; hybrid = sum of its components (p95 not defined).

System	Index (s)	Index (MB)	Query (ms)	p95 (ms)	Hardware
BM25	20.3	29.3	1.6	1.7	cpu
E5	174.6	116.0	17.3	18.2	Tesla T4
SPECTER2	182.6	116.0	30.0	36.6	Tesla T4
BM25+E5	194.9	145.4	19.3	–	–
BM25+SPECTER2	202.8	145.4	32.0	–	–

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